

Problem Set 3

Applied Stats II

Due: March 25, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in .pdf form.
- This problem set is due before 23:59 on Wednesday March 25, 2026. No late assignments will be accepted.

Question 1

We are interested in how governments' management of public resources impacts economic prosperity. Our data come from Alvarez, Cheibub, Limongi, and Przeworski (1996) and is labelled `gdpChange.csv` on GitHub. The dataset covers 135 countries observed between 1950 or the year of independence or the first year for which data on economic growth are available ("entry year"), and 1990 or the last year for which data on economic growth are available ("exit year"). The unit of analysis is a particular country during a particular year, for a total $> 3,500$ observations.

- Response variable:
 - `GDPWdiff`: Difference in GDP between year t and $t-1$. Possible categories include: "positive", "negative", or "no change"
- Explanatory variables:
 - `REG`: 1=Democracy; 0=Non-Democracy
 - `OIL`: 1=if the average ratio of fuel exports to total exports in 1984-86 exceeded 50%; 0= otherwise

Please answer the following questions:

1. Construct and interpret an unordered multinomial logit with GDPWdiff as the output and “no change” as the reference category, including the estimated cutoff points and coefficients.

```

1 # (1.1) Unordered multinomial logit
2 # change baseline category to "no change"
3 gdp_data$GDPWdiff <- factor(gdp_data$GDPWdiff, ordered = F)
4 gdp_data$GDPWdiff <- relevel(gdp_data$GDPWdiff, ref="no change")
5
6 # run unordered multinomial logit model
7 multinom_model <- multinom(GDPWdiff ~ REG + OIL, data = gdp_data)
8 summary(multinom_model)
9
10 # calculate z-scores and p-values "by hand" (since multinom does not
    provide p-values)
11 z_scores_multi <- summary(multinom_model)$coefficients / summary(multinom
    _model)$standard.errors
12 p_values_multi <- (1 - pnorm(abs(z_scores_multi), 0, 1)) * 2
13 p_values_multi

```

```

# weights:  12 (6 variable)
initial value 4087.936326
iter  10 value 2340.076844
final value 2339.385155
convergedtent...

```

Coefficients:

	(Intercept)	REG	OIL
negative	3.805370	1.379282	4.783968
positive	4.533759	1.769007	4.576321

Std. Errors:

	(Intercept)	REG	OIL
negative	0.2706832	0.7686958	6.885366
positive	0.2692006	0.7670366	6.885097

```
> p_values_multi
```

	(Intercept)	REG	OIL
negative	0 0.07276308	0.4871792	
positive	0 0.02109459	0.5062612	

We estimated an unordered multinomial logit model with “no change” as the reference category.

- **Comparing “negative” to “no change”:** The estimated intercept for the ‘negative’ versus ‘no change’ comparison is 3.805. Being a democracy (REG=1) increases the log-odds of experiencing negative GDP growth (compared to no change) by 1.379, holding OIL constant ($p = 0.073$).
- **Comparing “positive” to “no change”:** The estimated intercept for the ‘positive’ versus ‘no change’ comparison is 4.534. Being a democracy (REG=1) significantly increases the log-odds of experiencing positive GDP growth (compared to no change) by 1.769, holding OIL constant ($p = 0.021$).
- The OIL coefficient is positive in both comparisons, suggesting oil-exporting countries may have higher log-odds of being in either the negative or positive GDP-change category relative to ‘no change,’ but these estimates are not statistically significant.

Overall, democracy is clearly associated with higher log-odds of positive GDP change relative to no change. For negative GDP change relative to no change, the coefficient is also positive, but the evidence is weaker and not statistically significant at the 5% level.

2. Construct and interpret an ordered multinomial logit with GDPWdiff as the outcome variable, including the estimated cutoff points and coefficients.

```

1 # (1.2) Ordered multinomial logit
2 # Set ordered factor correctly: negative < no change < positive
3 gdp_data$GDPWdiff_ord <- factor(gdp_data$GDPWdiff, levels = c("negative",
4   "no change", "positive"), ordered = TRUE)
5
6 # run ordered multinomial logit model
7 ordered_model <- polr(GDPWdiff_ord ~ REG + OIL, data = gdp_data, Hess =
8   TRUE)
9 summary(ordered_model)
10
11 # calculate p-values for polr model
12 coef_table_ord <- coef(summary(ordered_model))
13 p_values_ord <- (1 - pnorm(abs(coef_table_ord[, "t value"]), 0, 1)) * 2
14 cbind(coef_table_ord, "p value" = p_values_ord)

```

	Value	Std. Error	t value	p value
REG	0.3984834	0.07518479	5.300054	1.157687e-07
OIL	-0.1987177	0.11571713	-1.717271	8.592967e-02
negative no change	-0.7311784	0.04760375	-15.359680	0.000000e+00
no change positive	-0.7104851	0.04750680	-14.955440	0.000000e+00

We estimated an ordered multinomial logit model with categories ordered as negative < no change < positive.

- **Cutoff points (Intercepts):** The estimated thresholds are -0.731 for the negative/no-change split and -0.710 for the no-change/positive split. The coefficient on REG is 0.398, meaning democracy increases the log-odds of being in a higher GDP-change category, holding OIL constant, and this effect is highly statistically significant ($p < 0.001$).
- **Coefficients:** The coefficient on OIL is -0.199, suggesting a lower likelihood of being in a higher GDP-change category, but this estimate is not statistically significant at the 5% level ($p = 0.086$).

Question 2

Consider the data set `MexicoMuniData.csv`, which includes municipal-level information from Mexico. The outcome of interest is the number of times the winning PAN presidential candidate in 2006 (`PAN.visits.06`) visited a district leading up to the 2009 federal elections, which is a count. Our main predictor of interest is whether the district was highly contested, or whether it was not (the PAN or their opponents have electoral security) in the previous federal elections during 2000 (`competitive.district`), which is binary (1=close/swing district, 0="safe seat"). We also include `marginality.06` (a measure of poverty) and `PAN.governor.06` (a dummy for whether the state has a PAN-affiliated governor) as additional control variables.

- (a) Run a Poisson regression because the outcome is a count variable. Is there evidence that PAN presidential candidates visit swing districts more? Provide a test statistic and p-value.

```

1 # (2a)
2 # run Poisson regression (outcome is count)
3 poisson_model <- glm(PAN.visits.06 ~ competitive.district + marginality
  .06 + PAN.governor.06, data=mexico_elections, family = poisson(link="
  log"))
4 summary(poisson_model)
5
6 # option #1: use info from summary() output (Wald test for competitive.
  district)
7 summary(poisson_model)$coefficients["competitive.district", ]
8
9 # option #2: manually run reduced model & anova (Likelihood Ratio Test)
10 poisson_reduced <- glm(PAN.visits.06 ~ marginality.06 + PAN.governor.06,
  data=mexico_elections, family = poisson(link="log"))
11 anova(poisson_reduced, poisson_model, test="LRT")

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.81023	0.22209	-17.156	<2e-16 ***
competitive.district	-0.08135	0.17069	-0.477	0.6336

```

marginality.06      -2.08014    0.11734 -17.728   <2e-16 ***
PAN.governor.06     -0.31158    0.16673  -1.869    0.0617 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for poisson family taken to be 1)

```

Null deviance: 1473.87  on 2406  degrees of freedom
Residual deviance:  991.25  on 2403  degrees of freedom
AIC: 1299.2

```

Number of Fisher Scoring iterations: 7

```

> summary(poisson_model)$coefficients["competitive.district", ]
      Estimate Std. Error      z value      Pr(>|z|)
-0.08135181  0.17068819 -0.47661062  0.63363942

> anova(poisson_reduced, poisson_model, test="LRT")
Analysis of Deviance Table
    Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         2404       991.48
2         2403       991.25  1   0.22377  0.6362

```

No. There is no statistically significant evidence that PAN presidential candidates visited swing districts more often. In fact, the estimated coefficient on competitive.district is negative (-0.081), implying a slightly lower expected visit count in competitive districts, but this effect is substantively small and statistically insignificant (Wald $z = -0.477$, $p = 0.634$; LRT $p = 0.636$).

(b) Interpret the marginality.06 and PAN.governor.06 coefficients.

```

1 # (2b)
2 # exponentiate coefficients to get Incidence Rate Ratios (IRR) for
  interpretation
3 round(exp(coef(poisson_model)), 3)

```

(Intercept)	competitive.district	marginality.06	PAN.governor.06
0.022	0.922	0.125	0.732

(Note: Exponentiating the Poisson coefficients yields Incidence Rate Ratios).

- **marginality.06:** For a one-unit increase in the poverty level, the expected number of PAN presidential visits changes by a **multiplicative factor of 0.125** ($e^{-2.080}$), holding other variables constant. In other words, visits drastically decrease as poverty increases.
- **PAN.governor.06:** The incidence rate ratio for PAN.governor.06 is 0.732, suggesting that districts in states with a PAN-affiliated governor are expected to receive about 26.8% fewer visits than districts in states without a PAN governor, holding other variables constant. However, this estimate is not statistically significant at the 5% level ($p = 0.0617$).

- (c) Provide the estimated mean number of visits from the winning PAN presidential candidate for a hypothetical district that was competitive (`competitive.district=1`), had an average poverty level (`marginality.06 = 0`), and a PAN governor (`PAN.governor.06=1`).

```
1 # (2c)
2 # option #1: predict function
3 round(predict(poisson_model, newdata = data.frame(competitive.district=1,
  marginality.06=0, PAN.governor.06=1), type="response"), 2)
```

```
1
0.01
```

```
1 # option #2: predicted count (exp(beta0 + beta1*X1 + ...))
2 exp_coefs_count <- exp(coef(poisson_model)[1] + (coef(poisson_model)[2] *
  1) + (coef(poisson_model)[3]*0) + (coef(poisson_model)[4]*1))
3 round(exp_coefs_count, 2)
```

```
(Intercept)
0.01
```

For this hypothetical district, the estimated mean number of visits from the winning PAN candidate is **0.01 visits** on average.

(Calculated using the inverse log link: $\exp(-3.810 - 0.081(1) - 2.080(0) - 0.312(1)) \approx 0.01$).